E. Scheme

time limit per test: 2 seconds
memory limit per test: 256 megabytes

To learn as soon as possible the latest news about their favourite fundamentally new operating system, BolgenOS community from Nizhni Tagil decided to develop a scheme. According to this scheme a community member, who is the first to learn the news, calls some other member, the latter, in his turn, calls some third member, and so on; i.e. a person with index i got a person with index f_i, to whom he has to call, if he learns the news. With time BolgenOS community members understood that their scheme doesn't work sometimes — there were cases when some members didn't learn the news at all. Now they want to supplement the scheme: they add into the scheme some instructions of type (x_i, y_i), which mean that person x_i has to call person y_i as well. What is the minimum amount of instructions that they need to add so, that at the end everyone learns the news, no matter who is the first to learn it?

Input

The first input line contains number n $(2 \le n \le 10^5)$ — amount of BolgenOS community members. The second line contains n space-separated integer numbers f_i $(1 \le f_i \le n, i \ne f_i)$ — index of a person, to whom calls a person with index i.

Output

In the first line output one number — the minimum amount of instructions to add. Then output one of the possible variants to add these instructions into the scheme, one instruction in each line. If the solution is not unique, output any.

Examples

input Copy

```
3
3 3 2
```

output Copy

```
1
3 1
```

input Copy

```
7
2 3 1 3 4 4 1
```

output Copy

```
3
2 5
2 6
3 7
```

→ Attention

The package for this problem was not updated by the problem writer or Codeforces administration after we've upgraded the judging servers. To adjust the time limit constraint, a solution execution time will be multiplied by 2. For example, if your solution works for 400 ms on judging servers, then the value 800 ms will be displayed and used to determine the verdict.

Codeforces Beta Round 22 (Div. 2 Only)

Finished

Practice

→ Virtual participation

Virtual contest is a way to take part in past contest, as close as possible to participation on time. It is supported only ICPC mode for virtual contests. If you've seen these problems, a virtual contest is not for you - solve these problems in the archive. If you just want to solve some problem from a contest, a virtual contest is not for you - solve this problem in the archive. Never use someone else's code, read the tutorials or communicate with other person during a virtual contest.

Start virtual contest

→ Clone Contest to Mashup

You can clone this contest to a mashup.

Clone Contest

→ Submit?

Language: GNU G++23 14.2 (64 bit, msys2

Choose file: Choose File No file chosen

Submit

→ Last submissions

Submission	Time	Verdict
371748071	Apr/19/2026 12:28	Accepted
371747853	Apr/19/2026 12:26	Accepted
371746918	Apr/19/2026 12:15	Wrong answer on test 3

371746668	Apr/19/2026 12:13	Wrong answer on test 3
371746424	Apr/19/2026 12:10	Wrong answer on test 3
371746167	Apr/19/2026 12:07	**Accepted**
371746019	Apr/19/2026 12:06	Memory limit exceeded on test 1
371745784	Apr/19/2026 12:03	Memory limit exceeded on test 1
371745521	Apr/19/2026 12:00	Memory limit exceeded on test 1
371745198	Apr/19/2026 11:57	Runtime error on test 1

→ Problem tags

dfs and similar | graphs | trees | *2300

No tag edit access

→ Contest materials

- Announcement
- Tutorial

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